IPFS (InterPlanetary File System) – Decentralized Storage for Blockchain

What is IPFS?

IPFS (InterPlanetary File System) is a decentralized peer-to-peer storage network that allows files to be stored and shared across multiple computers instead of a single centralized server.

It was developed by the organization Protocol Labs.

Think of IPFS as:

- Traditional Web = "Where is the file?" (Location-based)
- IPFS = "What is the file?" (Content-based)

Instead of finding a file using a server address, IPFS finds a file using its unique cryptographic hash.

Why Do We Need IPFS in Blockchain?

Blockchains such as Ethereum are not designed to store large files because:

Problem	Explanation
Expensive	Storing data on-chain costs gas fees
Slow	Large files increase blockchain size
Scalability Issues	Every node stores the same data
Inefficient	Images, videos, PDFs are too large

Solution ->

Store:

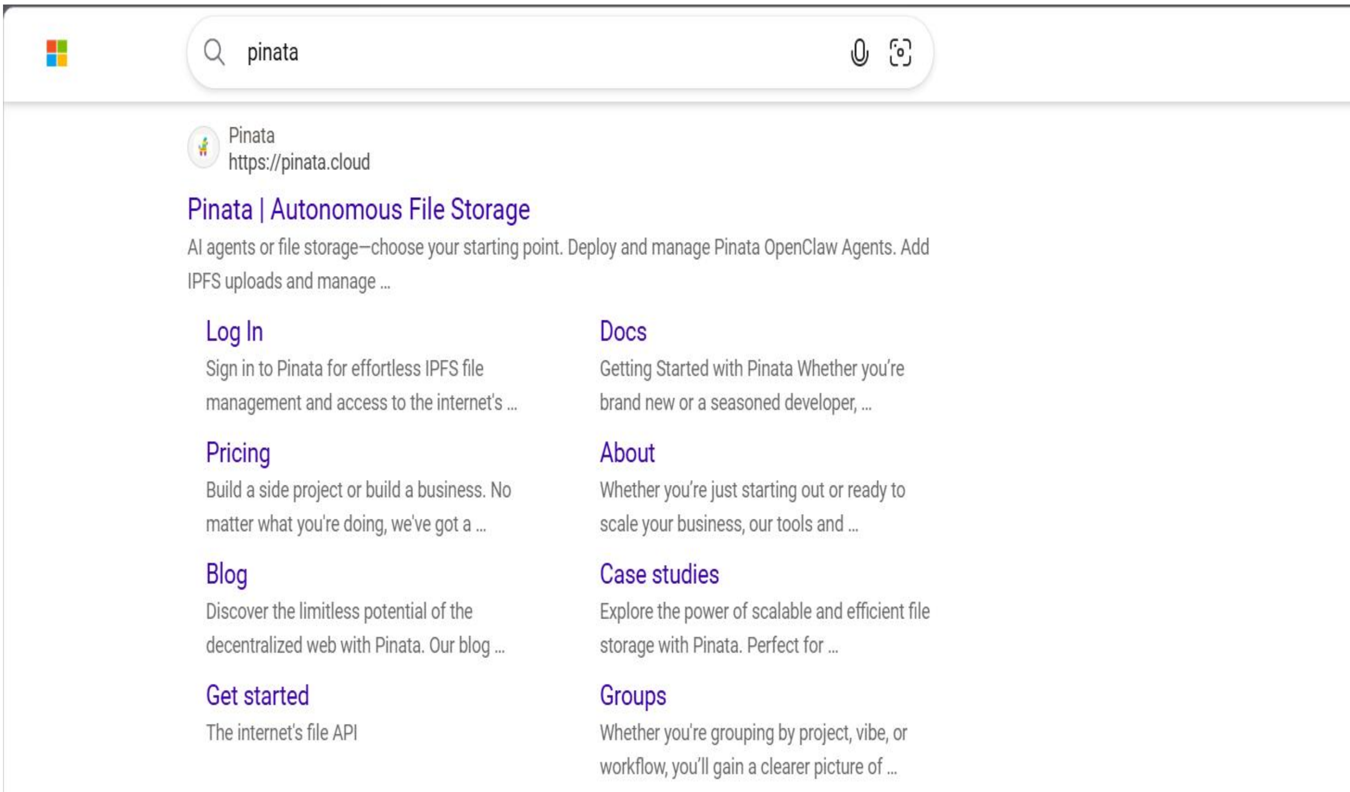
- File → IPFS
- File Hash (CID) → Blockchain

This keeps blockchain storage cheap and efficient.

How IPFS Works ->

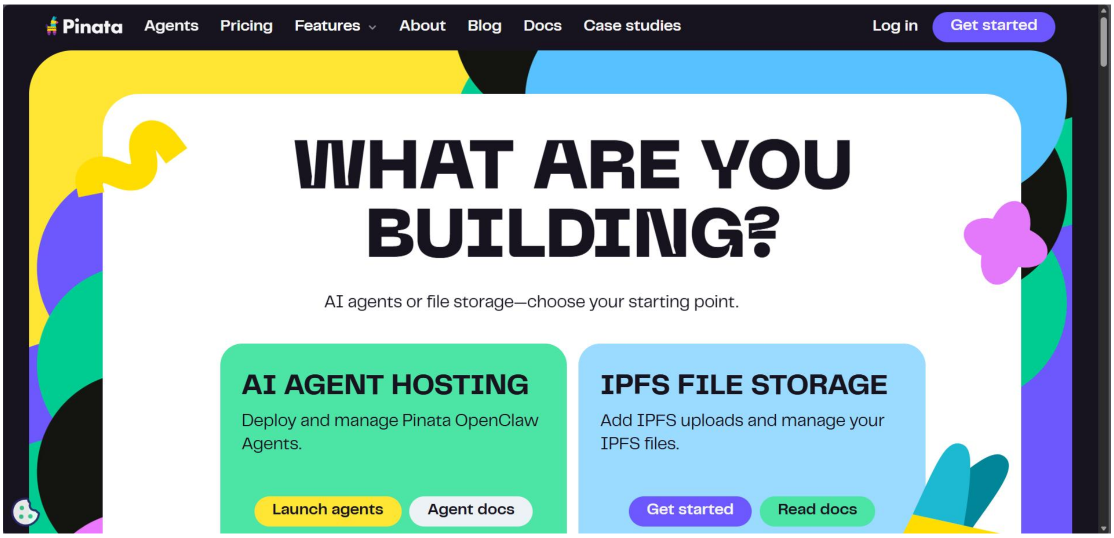
Step1: Register on IPFS website (e.g Pinata)

Go to website

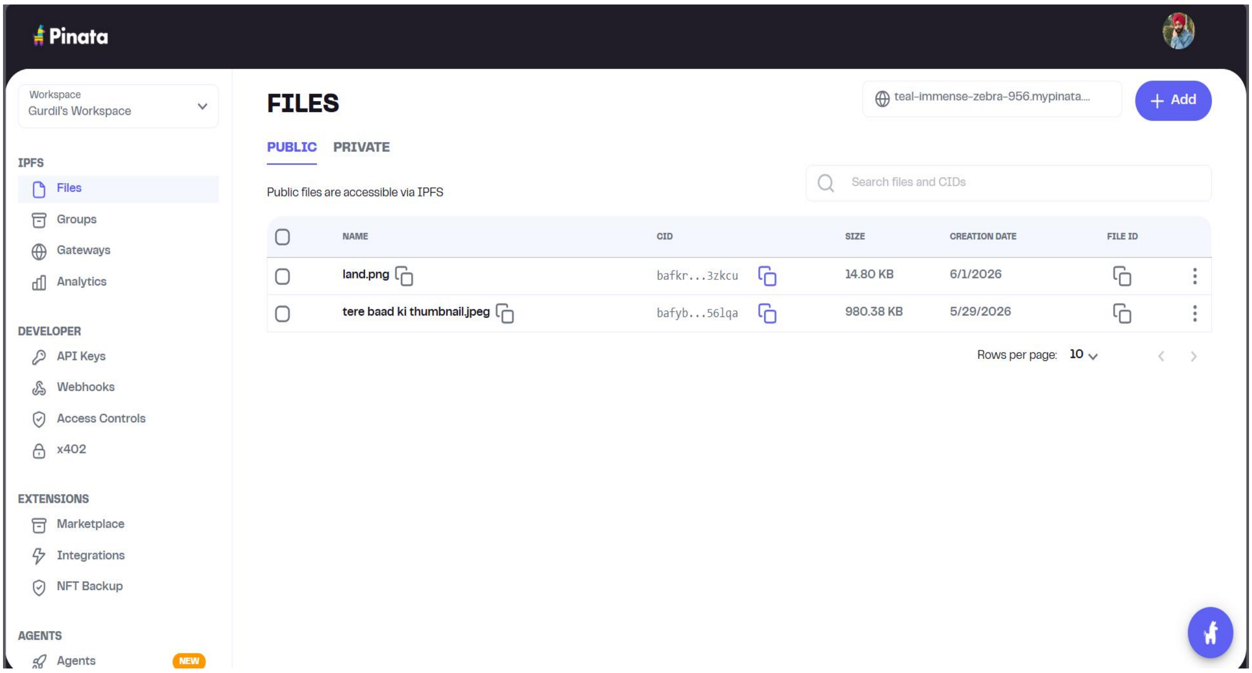


Open the first link.

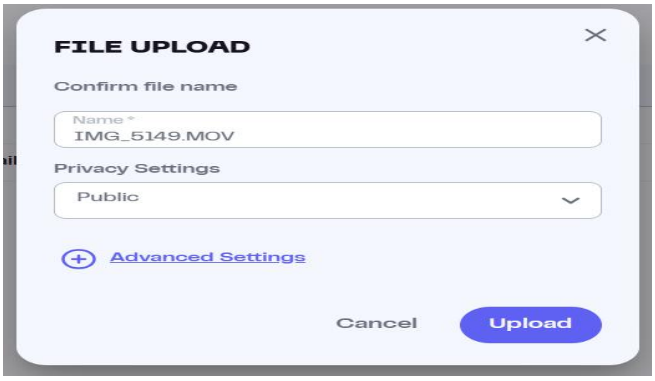
<u>Pinata</u> looks like ->



After signing in to <u>Pinata</u> we have an interface which allow us to upload files to IPFS

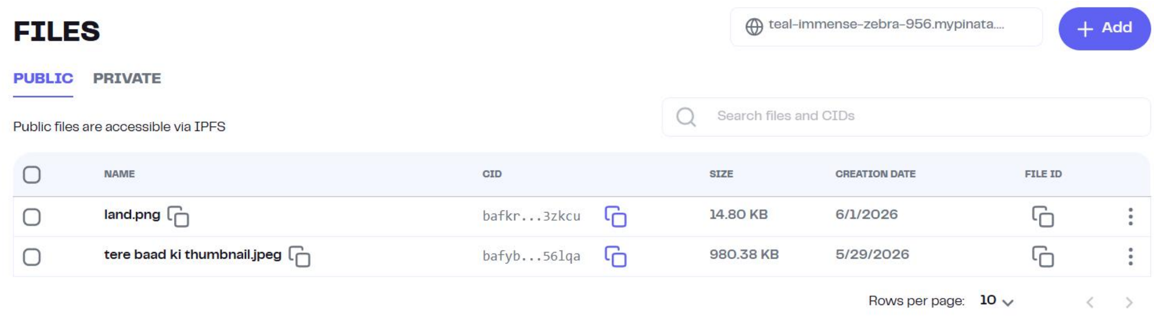


As we see I have already uploaded 2 files here. We can add new files by clicking on Add button on the top right of the site after clicking on it we have to choose file and after selecting file we have:



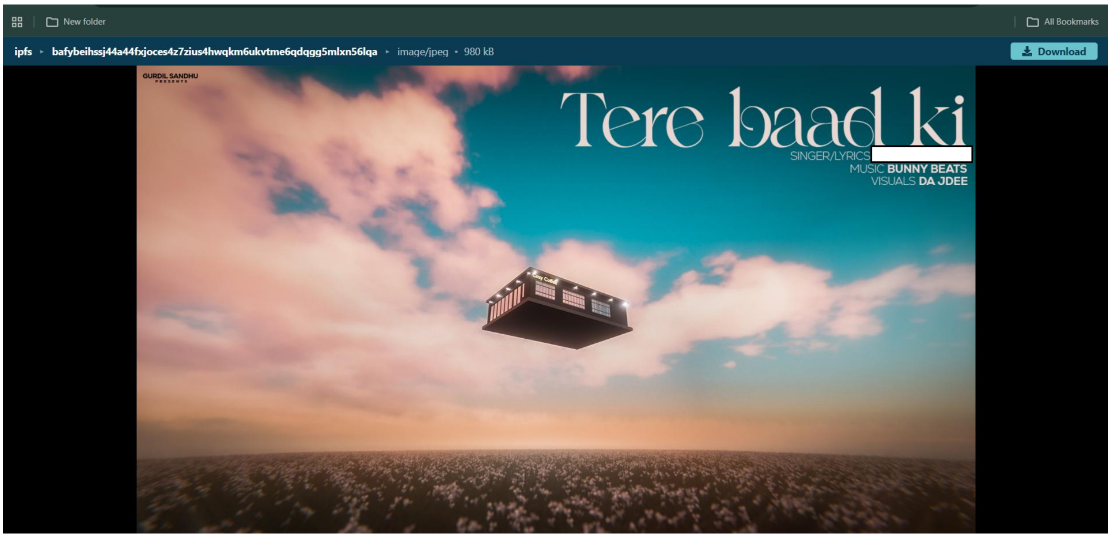
We can change the file name and set it as per our choice and for privacy settings Pinata gives us option for public or private we can choose as per our needs.

After uploading file file will be shown on either public or private section:



Now as we see file has properties like NAME, CID, SIZE, CREATION DATE, FILE ID.
So what is CID?, CID we can say Content Identifier. It is used to find files on IPFS
So here in IPFS files are find by Content(CID), not by address. Also CID changes according to the data or content file has even the slightest change make does effect.

If the file uploaded is public then anyone with CID can see the file. To do so copy the CID. Now there is a url which is used (https://ipfs.io/ipfs/CID) on the place of CID past the copied CID from Pinata. On searching on any browser the file uploaded will be shown like:



This file is uploaded by me as JPEG format. We can upload there any kind of file it can be video or any other unstructured data.

Relationship Between Blockchain and IPFS:

Without IPFS->

Blockchain
└── Image
└── Video
└── PDF

There are problems doing so:
1. Huge Chain size.
2. High Gas Fees.

With IPFS->

IPFS
└── Image
└── Video
└── PDF

Blockchain
└── CID only

Now Advantages doing this:
1. Lower Cost.
2. Faster Transactions.
3. Better scalability.

IPFS using Smart Contract:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract IPFS{
    string cId;

    function addCid(string memory _cId)public {
        cId = _cId;
    }
    function geturl()public view returns(string memory){
        string memory mainURL = string(abi.encodePacked("https://ipfs.io/ipfs/",cId));

        return mainURL;
    }
}

```

What this code say?

1. Contract named as IPFS.
2. CID variable was build using string datatype named cId.
3. Then function named addCid where user add the CID hash like we saw earlier from pinata.
4. CID will be stored in state variable.
5. Second function named geturl is to generate IPFS url. So this function returns mainUrl or we can say IPFS url like https://ipfs.io/ipfs/CID